Computer Networks

Lecture 20 Network Layer

Dr. Tauheed Ahmed

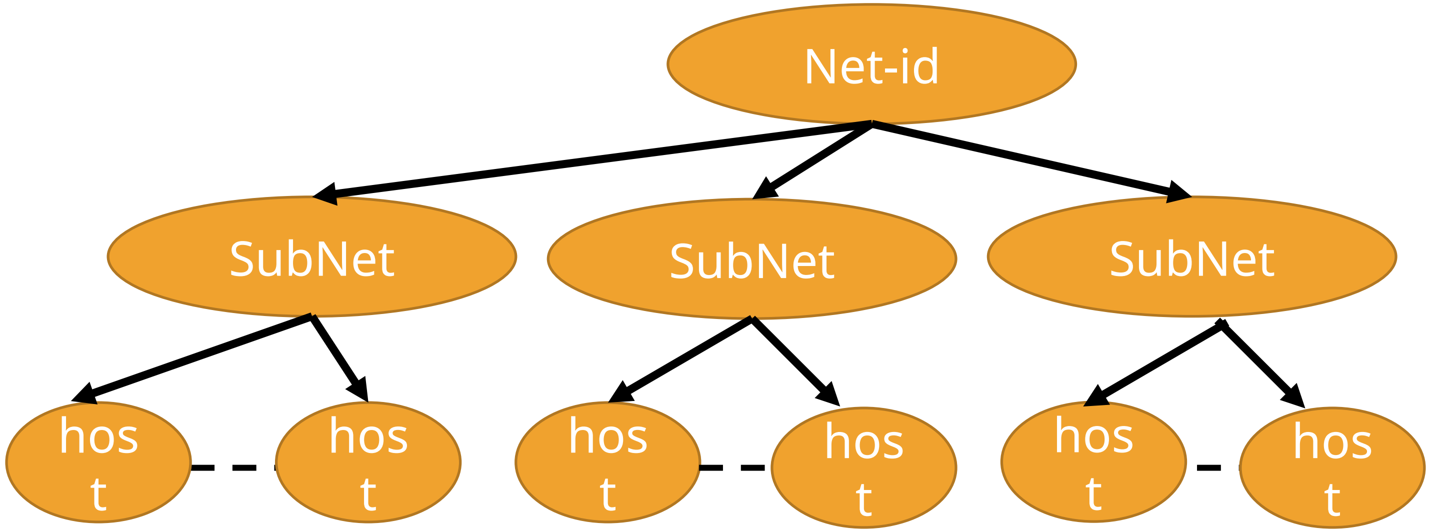
Content

- **Subnetting**

Subnetting

During the era of classful addressing, subnetting was introduced.

If an organization was granted a large block in class *A or B*, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets).



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Questions

1. If subnet mask of a class B is 255.255.240.0. Calculate no. of subnets and no. of host in each subnet.
2. If subnet mask of a class C is 255.255.255.224. Calculate no. of subnets and no. of host in each subnet.
3. If subnet mask of a class A is 255.255.192.0. Calculate no. of subnets and no. of host in each subnet.
4. IP address is 197.111.121.199 and subnet mask is 255.255.255.240. Calculate subnet id, last subnet id, 1st subnet id 2nd subnet id, last host of last subnet.
5. IP address is 203.112.11.117 and subnet mask 255.255.255.224. Find subnet id to which this ip belong
   - IP => 203.112.111.96
6. In the previous question calculate the DBA of 4th subnet and last subnet
7. IP address of the system is 61.119.189.176 and the subnet mask is 255.255.192.0
   - Calculate the DBA of 1st subnet
   - Calculate last host of last subnet
8. If IP address of the system S_1 is 194.112.116.67 and IP address of another System S_2 is 194.112.116.84 and subnet mask is 255.255.255.240.
   - Identify weather the hosts belong to same subnet or not.
   - Mention their subnet id no.